

Keon Systems

The Constitutional AI Execution Substrate

Version 2.0
Status: Canonical Architecture

This document defines Cryptographically Governed AI Execution (CGAE) and the architectural invariants enforced by the Keon Systems substrate.

Runtime Source of Truth: keon-systems/docs/governance/.md and keon-systems/launch/keon-v-launch-manifest.yaml*

1. The Problem: Trust-Me Architecture

Traditional AI agent frameworks execute before authorization and rely on post-execution logging. If logging fails, the system cannot prove what occurred. Divergence between side effects and audit trails creates legal and operational liability.

Keon makes out-of-policy execution structurally impossible before it begins.

2. The Standard: Cryptographically Governed AI Execution (CGAE)

Invariant	Statement
I-EXEC-01	Execution SHALL NOT proceed unless authorization receipt is durably persisted prior to side effects.
I-MEM-01	Memory mutation SHALL NOT occur without corresponding persisted memory receipt.
I-LEDGER-01	Ledger entries SHALL be append-only and immutable within a partition.
I-VERIFY-01	All governance artifacts MUST be verifiable offline independent of runtime.

3. The Substrate: ARO, JCS, and Keon Memory

The Authoritative Receipt Outbox (ARO) enforces receipt-gated state mutation. All governed state transitions route through ARO and fail closed.

Pending → Persisted → Verified → Applied → Completed
■
Failed (terminal)

Keon applies RFC 8785 JSON Canonicalization Scheme (JCS) before hashing. Receipts are signed using Ed25519 and verified offline.

Keon Memory is an append-only, UUIDv7-ordered, partition-scoped constitutional ledger. Receipts, traces, and evidence are ledger-derived.

4. ALPHA Authority Doctrine

Letter	Meaning
A	Attested — cryptographically signed, canonicalized, non-repudiable
L	Ledger — append-only causal spine (Keon Memory)
P	Policy-Bound — tied to deterministic PolicyHash
H	Human — explicit authorization domains
A	Authority — binding governance transitions

No authority. No execution. No exceptions.

5. Governed Execution (GE-001)

Keon exposes governance through MCP tools including `keon.governed.execute`, `keon.runs.get`, `keon.runs.stream`, `keon.evidence.export`, and `keon.evidence.verify`.

QUEUED → DECIDING → BINDING → EXECUTING → FINALIZING → COMPLETED
■ DENIED / FAILED

6. Evidence Packs

A completed governed run produces an Evidence Pack containing signed receipts, policy context, and Spine references. Packs are verifiable offline using `keon-cli`.

```
keon-cli verify-pack --path evidence-pack.zip  
# Tamper detection produces non-zero exit codes
```

If the Evidence Pack verifies, the execution was governed. If it does not verify, it was not. There is no third state.

7. Constitutional Trust Model

Keon does not require trust in its runtime. It requires only verification of output. Receipts are cryptographically bound, ledger-derived, and independently auditable.

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